

DOCKER-02

1.Create a tomcat container on 8080 and deploy sample application in tomcat.

Lets pull tomcat image

docker pull tomcat:latest

verify the image

docker images

```
root@ip-172-31-24-106: ~  
root@ip-172-31-24-106:~# docker pull tomcat:latest  
latest: Pulling from library/tomcat  
4f4fb700ef54: Pull complete  
65313903a8a8: Pull complete  
6b82a07ff6e7: Pull complete  
13883d0c1f97: Download complete  
f80ef884f8c2: Download complete  
Digest: sha256:247650787b97751d68aa2d532d6e09111c36946d9a50ea0e5071e19facc1e5e7  
Status: Downloaded newer image for tomcat:latest  
docker.io/library/tomcat:latest  
root@ip-172-31-24-106:~# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
jenkins/jenkins:latest	01c992ffef29	799MB	289MB	
nginx:latest	5aca99593157	241MB	66MB	
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB	
tomcat:latest	247650787b97	589MB	158MB	
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	

```
root@ip-172-31-24-106:~# |
```

Create and Start the Tomcat Container

- Run the Tomcat container and expose it on port 8080.

```
root@ip-172-31-24-106:/home/ubuntu# docker run -d --name tomcat-server -p 8080:8080 tomcat:latest  
a7e5a41f5fe83777946b55555241293e21ddcb86eadcef545a186f973166f7d4  
root@ip-172-31-24-106:/home/ubuntu# |
```

Verify the container is running or not

```
root@ip-172-31-24-106:/home/ubuntu# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
a7e5a41f5fe8	tomcat:latest	"catalina.sh run"	9 minutes ago	Up 9 minutes

```
root@ip-172-31-24-106:/home/ubuntu# |
```

→Login to the running Tomcat container

docker exec -it tomcat-server bash

→ Navigate to the Tomcat deployment directory

cd /usr/local/tomcat/webapps

pwd

→ Update package repositories inside the container:

apt update

```
root@ip-172-31-24-106:/home/ubuntu# docker exec -it tomcat-server bash
root@a7e5a41f5fe8:/usr/local/tomcat# cd /usr/local/tomcat/webapps
root@a7e5a41f5fe8:/usr/local/tomcat/webapps# pwd
/usr/local/tomcat/webapps
root@a7e5a41f5fe8:/usr/local/tomcat/webapps# apt update
Get:1 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [48.4 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,511 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [2,083 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [3,683 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1,808 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [4,013 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [2,083 kB]
```

```

Get:9 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1,808 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [4,013 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,154 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2,500 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [54.6 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [490 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [35.9 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [671 B]
Fetched 38.3 MB in 2s (25.5 MB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
6 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@a7e5a41f5fe8:/usr/local/tomcat/webapps#

```

apt install wget -y

```

root@a7e5a41f5fe8:/usr/local/tomcat/webapps# apt install wget -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libpsl5t64 publicsuffix
The following NEW packages will be installed:
  libpsl5t64 publicsuffix wget
0 upgraded, 3 newly installed, 0 to remove and 6 not upgraded.
Need to get 520 kB of archives.
After this operation, 1,423 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 libpsl5t64 amd64 0.21.2-1.1build1 [57.1 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 publicsuffix all 20231001.0357-0.1 [129 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 wget amd64 1.21.4-1ubuntu4.1 [334 kB]
Fetched 520 kB in 0s (9,881 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libpsl5t64:amd64.
(Reading database ... 6571 files and directories currently installed.)
Preparing to unpack .../libpsl5t64_0.21.2-1.1build1_amd64.deb ...
Unpacking libpsl5t64:amd64 (0.21.2-1.1build1) ...
Selecting previously unselected package publicsuffix.

```

→Download Sample WAR File

wget <https://tomcat.apache.org/tomcat-7.0-doc/appdev/sample/sample.war>


```

root@a7e5a41f5fe8:/usr/local/tomcat/webapps# wget https://tomcat.apache.org/tomcat-7.0-doc/appdev/sample/sample.war
--2026-06-01 19:23:09-- https://tomcat.apache.org/tomcat-7.0-doc/appdev/sample/sample.war
Resolving tomcat.apache.org (tomcat.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to tomcat.apache.org (tomcat.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4606 (4.5K)
Saving to: 'sample.war'

sample.war          100%[=====>]    4.50K  --.-KB/s    in 0s

2026-06-01 19:23:09 (103 MB/s) - 'sample.war' saved [4606/4606]

root@a7e5a41f5fe8:/usr/local/tomcat/webapps#

```

```

root@a7e5a41f5fe8:/usr/local/tomcat/webapps# ls
sample  sample.war
root@a7e5a41f5fe8:/usr/local/tomcat/webapps# exit
exit
root@ip-172-31-24-106:/home/ubuntu#

```

verify Application Deployment Using Docker Logs

docker logs tomcat-server

```

root@ip-172-31-24-106:/home/ubuntu# docker logs tomcat-server
01-Jun-2026 19:04:12.382 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log Server version name: Apache Tomcat/11.0.22
01-Jun-2026 19:04:12.387 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log Server built: May 1 2026 18:29:05 UTC
01-Jun-2026 19:04:12.387 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log Server version number: 11.0.22.0
01-Jun-2026 19:04:12.389 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log OS Name: Linux
01-Jun-2026 19:04:12.389 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log OS Version: 7.0.0-1004-aws
01-Jun-2026 19:04:12.389 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log Architecture: amd64
01-Jun-2026 19:04:12.389 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log Java Home: /opt/java/openjdk
01-Jun-2026 19:04:12.390 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log JVM Version: 25.0.3+9-LTS
01-Jun-2026 19:04:12.390 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log JVM Vendor: Eclipse Adoptium
01-Jun-2026 19:04:12.390 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log CATALINA_BASE: /usr/local/tomcat
01-Jun-2026 19:04:12.390 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log CATALINA_HOME: /usr/local/tomcat
01-Jun-2026 19:04:12.420 INFO [main] org.apache.catalina.startup.VersionLoggerLi

```

```
root@ip-172-31-24-106: /home/ubuntu
01-Jun-2026 19:04:12.451 INFO [main] org.apache.catalina.core.AprLifecycleListener.lifecycleEvent Loaded Apache Tomcat Native library [2.0.14] using APR version [1.7.2].
01-Jun-2026 19:04:12.506 INFO [main] org.apache.catalina.core.AprLifecycleListener.initializeSSL OpenSSL successfully initialized [OpenSSL 3.0.13 30 Jan 2024]
01-Jun-2026 19:04:12.934 INFO [main] org.apache.coyote.AbstractProtocol.init Initializing ProtocolHandler ["http-nio-8080"]
01-Jun-2026 19:04:12.984 INFO [main] org.apache.catalina.startup.Catalina.load Server initialization in [959] milliseconds
01-Jun-2026 19:04:13.051 INFO [main] org.apache.catalina.core.StandardService.startInternal Starting service [Catalina]
01-Jun-2026 19:04:13.052 INFO [main] org.apache.catalina.core.StandardEngine.startInternal Starting Servlet engine: [Apache Tomcat/11.0.22]
01-Jun-2026 19:04:13.068 INFO [main] org.apache.coyote.AbstractProtocol.start Starting ProtocolHandler ["http-nio-8080"]
01-Jun-2026 19:04:13.083 INFO [main] org.apache.catalina.startup.Catalina.start Server startup in [96] milliseconds
01-Jun-2026 19:23:13.182 INFO [Catalina-utility-1] org.apache.catalina.startup.HostConfig.deployWAR Deploying web application archive [/usr/local/tomcat/webapps/sample.war]
01-Jun-2026 19:23:13.592 INFO [Catalina-utility-1] org.apache.catalina.startup.HostConfig.deployWAR Deployment of web application archive [/usr/local/tomcat/webapps/sample.war] has finished in [407] ms
root@ip-172-31-24-106: /home/ubuntu#
```

docker ps

docker exec -it tomcat-server bash

cd /usr/local/tomcat/webapps

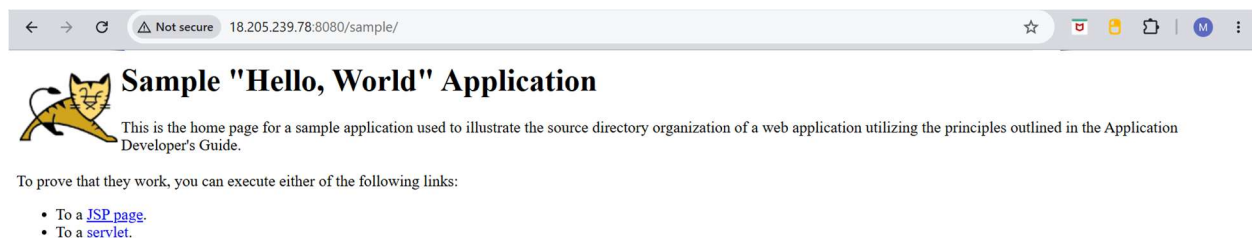
ls -lrt

exit

curl -I http://localhost:8080/sample

```
root@ip-172-31-24-106: /home/ubuntu
root@ip-172-31-24-106:/home/ubuntu# docker exec -it tomcat-server bash
root@a7e5a41f5fe8:/usr/local/tomcat# ls -lrt
total 232
drwxr-xr-x 7 root root 4096 May 1 18:29 webapps.dist
-rw-r--r-- 1 root root 16114 May 1 18:29 RUNNING.txt
-rw-r--r-- 1 root root 6470 May 1 18:29 RELEASE-NOTES
-rw-r--r-- 1 root root 3224 May 1 18:29 README.md
-rw-r--r-- 1 root root 2333 May 1 18:29 NOTICE
-rw-r--r-- 1 root root 60517 May 1 18:29 LICENSE
-rw-r--r-- 1 root root 8626 May 1 18:29 CONTRIBUTING.md
-rw-r--r-- 1 root root 26826 May 1 18:29 BUILDING.txt
-rw-r--r-- 1 root root 32010 May 5 23:13 upstream-KEYS
drwxrwxrwt 2 root root 4096 May 5 23:13 temp
drwxr-xr-x 2 root root 4096 May 5 23:13 lib
-rw-r--r-- 1 root root 30936 May 5 23:13 filtered-KEYS
drwxr-xr-x 2 root root 4096 May 5 23:13 native-jni-lib
drwxr-xr-x 2 root root 4096 May 5 23:14 bin
drwxr-xr-x 1 root root 4096 Jun 1 19:04 conf
drwxrwxrwt 1 root root 4096 Jun 1 19:04 logs
drwxr-xr-x 1 root root 4096 Jun 1 19:23 webapps
drwxrwxrwt 1 root root 4096 Jun 1 19:23 work
root@a7e5a41f5fe8:/usr/local/tomcat# exit
exit
root@ip-172-31-24-106:/home/ubuntu# curl -I http://localhost:8080/sample
HTTP/1.1 302
Location: /sample/
Date: Mon, 01 Jun 2026 19:31:46 GMT
root@ip-172-31-24-106:/home/ubuntu#
```

→ Successfully created a Tomcat Docker container on port 8080, deployed the sample.war application.



2.Create volume and deploy tomcat container on port 8081.

→ Create Docker Volume

```
root@ip-172-31-24-106:/home/ubuntu# docker volume create deployment
deployment
root@ip-172-31-24-106:/home/ubuntu# docker volume ls
DRIVER      VOLUME NAME
local       deployment
local       f609f28403f6919ab4190586746189938dad5df8adc8222e5a5cfaae1a6be952
root@ip-172-31-24-106:/home/ubuntu#
```

→ Verify Volume


```

root@ip-172-31-24-106:/home/ubuntu# docker volume ls
DRIVER      VOLUME NAME
local       deployment
local       f609f28403f6919ab4190586746189938dad5df8adc8222e5a5cfaae1a6be952
root@ip-172-31-24-106:/home/ubuntu#

```

→ Check Volume Location

docker volume inspect deployment

```

root@ip-172-31-24-106:/home/ubuntu# docker volume inspect deployment
[
  {
    "CreatedAt": "2026-06-01T19:37:48Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/deployment/_data",
    "Name": "deployment",
    "Options": null,
    "Scope": "local"
  }
]
root@ip-172-31-24-106:/home/ubuntu#

```

Download Sample WAR File into Volume

cd /var/lib/docker/volumes/deployment/_data

wget https://tomcat.apache.org/tomcat-7.0-doc/appdev/sample/sample.war

ls -lrt

```

root@ip-172-31-24-106:/home/ubuntu# cd /var/lib/docker/volumes/deployment/_data
root@ip-172-31-24-106:/var/lib/docker/volumes/deployment/_data# wget https://tomcat.apache.org/tomcat-7.0-doc/appdev/sample/sample.war
--2026-06-01 19:41:50-- https://tomcat.apache.org/tomcat-7.0-doc/appdev/sample/sample.war
Resolving tomcat.apache.org (tomcat.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to tomcat.apache.org (tomcat.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4606 (4.5K)
Saving to: 'sample.war'

sample.war      100%[=====>]    4.50K  --.-KB/s   in 0s
2026-06-01 19:41:50 (60.8 MB/s) - 'sample.war' saved [4606/4606]

root@ip-172-31-24-106:/var/lib/docker/volumes/deployment/_data# ls -lrt
total 8
-rw-r--r-- 1 root root 4606 Mar 31  2012 sample.war
root@ip-172-31-24-106:/var/lib/docker/volumes/deployment/_data#

```

Deploy Tomcat Container Using the Volume

**docker run -d **

**--name tomcat-volume **

**-p 8081:8080 **

-v

/var/lib/docker/volumes/deployment/_data:/usr/l

ocal/tomcat/webapps \

tomcat:latest

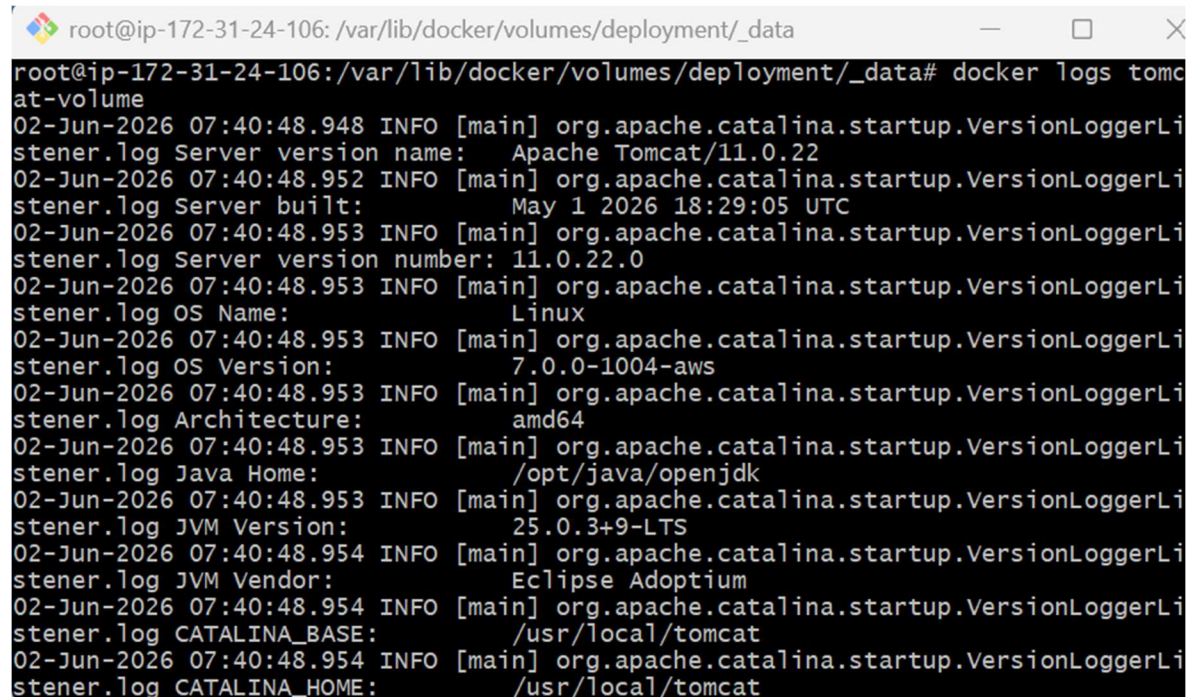
```
root@ip-172-31-24-106:/var/lib/docker/volumes/deployment/_data# docker run -d \
--name tomcat-volume \
-p 8081:8080 \
-v /var/lib/docker/volumes/deployment/_data:/usr/local/tomcat/webapps \
tomcat:latest
1f71a650f76207d8f68d3c36754696227d1ae71e2e25adf94f0174c18ec73473
root@ip-172-31-24-106:/var/lib/docker/volumes/deployment/_data#
```

Docker ps

```
root@ip-172-31-24-106:/var/lib/docker/volumes/deployment/_data# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS
PORTS         NAMES
1f71a650f762   tomcat:latest  "catalina.sh run"        32 seconds ago Up 31 second
s  0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp tomcat-volume
root@ip-172-31-24-106:/var/lib/docker/volumes/deployment/_data#
```

→Verify Deployment Logs

docker logs tomcat-volume

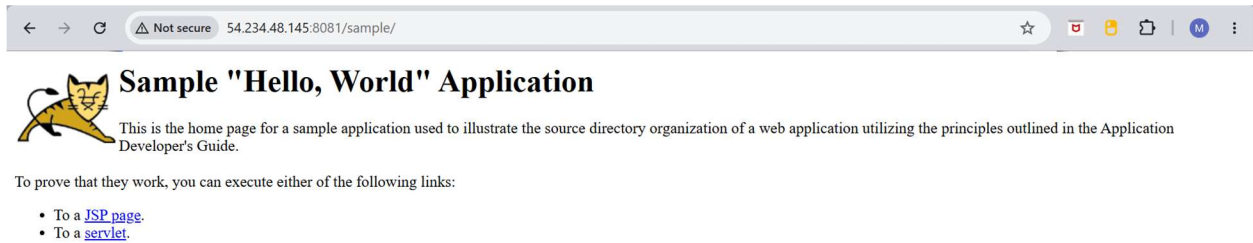


```
root@ip-172-31-24-106:/var/lib/docker/volumes/deployment/_data# docker logs tomcat-volume
02-Jun-2026 07:40:48.948 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log Server version name: Apache Tomcat/11.0.22
02-Jun-2026 07:40:48.952 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log Server built: May 1 2026 18:29:05 UTC
02-Jun-2026 07:40:48.953 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log Server version number: 11.0.22.0
02-Jun-2026 07:40:48.953 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log OS Name: Linux
02-Jun-2026 07:40:48.953 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log OS Version: 7.0.0-1004-aws
02-Jun-2026 07:40:48.953 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log Architecture: amd64
02-Jun-2026 07:40:48.953 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log Java Home: /opt/java/openjdk
02-Jun-2026 07:40:48.953 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log JVM Version: 25.0.3+9-LTS
02-Jun-2026 07:40:48.954 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log JVM Vendor: Eclipse Adoptium
02-Jun-2026 07:40:48.954 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log CATALINA_BASE: /usr/local/tomcat
02-Jun-2026 07:40:48.954 INFO [main] org.apache.catalina.startup.VersionLoggerLi
stener.log CATALINA_HOME: /usr/local/tomcat
```

curl -I http://localhost:8081/sample


```
root@ip-172-31-24-106:/var/lib/docker/volumes/deployment/_data# curl -I http://localhost:8081/sample
HTTP/1.1 302
Location: /sample/
Date: Tue, 02 Jun 2026 07:43:39 GMT
root@ip-172-31-24-106:/var/lib/docker/volumes/deployment/_data# |
```

→ Successfully created a Docker volume and deployed a Tomcat container on port 8081.



3.Limit the nginx container to 500 MB.

→Pull Nginx Image

docker pull nginx:latest

docker images

```
root@ip-172-31-24-106:/home/ubuntu# docker pull nginx:latest
latest: Pulling from library/nginx
Digest: sha256:5aca99593157f4ae539a5dec1092a0ad8762f8e2eb1789085a13a0f5622369f6
Status: Image is up to date for nginx:latest
docker.io/library/nginx:latest
root@ip-172-31-24-106:/home/ubuntu# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA	In Us
jenkins/jenkins:lts	01c992ffef29	799MB	289MB		
nginx:latest	5aca99593157	241MB	66MB		
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB		
tomcat:latest	247650787b97	589MB	158MB		
ubuntu:latest	f3d28607ddd7	160MB	45.3MB		

→Run Nginx Container with Memory Limit

**docker run -d **

**--name nginx-limit **

**-p 8082:80 **

**--memory=500m **

nginx:latest

```
root@ip-172-31-24-106:/home/ubuntu# docker run -d \
--name nginx-limit \
-p 8082:80 \
--memory=500m \
nginx:latest
015d0be59d7b34904405d6cbd20ccf96c0b86838b7adb95b121d120444dff5ea
root@ip-172-31-24-106:/home/ubuntu#
```

docker ps

```
root@ip-172-31-24-106:/home/ubuntu# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
015d0be59d7b	nginx:latest	"/docker-entrypoint...."	23 seconds ago	Up 23 s
1f71a650f762	tomcat:latest	"catalina.sh run"	10 minutes ago	Up 10 m

```
root@ip-172-31-24-106:/home/ubuntu#
```

→verified the limit using the docker stats command.

docker stats

```
root@ip-172-31-24-106:/home/ubuntu
```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
015d0be59d7b	nginx-limit	0.00%	14.63MiB / 500MiB	2.93%	866B / 126B	13.4MB / 12.3kB	3
1f71a650f762	tomcat-volume	0.15%	141.3MiB / 3.722GiB	3.71%	6.64kB / 6.3kB	64.1MB / 283kB	35

→Created an Nginx container with a memory limit of 500 MB

```
← → ↻ ⚠ Not secure 54.234.48.145:8082 ☆ 📄 🗑️ | M ⋮
```

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

4.Create a sample docker file using below

instructions.

1.Base module as amazonlinux:latest

2.Maintainer you name

3.Install nginx

4.COPY one index.html file to image

5.Expose on port 80

6.Command to start the nginx container

→Create index.html

Vi index.html

```
root@ip-172-31-24-106: /home/ubuntu
<html>
<head>
  <title>Docker Demo</title>
</head>
<body>
  <h1>Welcome to Nginx Docker Container!!</h1>
</body>
</html>
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
-- INSERT --
```

Vi dockerfile

```
root@ip-172-31-24-106: /home/ubuntu
FROM amazonlinux:latest
LABEL maintainer="Manisha Reddy"
RUN yum update -y && \
    yum install nginx -y
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
-- INSERT --
```

Build Docker Image

docker build -t nginx-custom:v1 .


```
root@ip-172-31-24-106: /home/ubuntu
root@ip-172-31-24-106:/home/ubuntu# docker build -t nginx-custom:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 14.85kB
Step 1/6 : FROM amazonlinux:latest
latest: Pulling from library/amazonlinux
c7345fb8c651: Pulling fs layer
b3d837a98232: Download complete
2da5356a79d8: Download complete
c7345fb8c651: Download complete
c7345fb8c651: Pull complete
Digest: sha256:ad882807c6c88f478186fa2632637380e06071a303e2b3bd96ab71846a77c6a3
Status: Downloaded newer image for amazonlinux:latest
----> ad882807c6c8
Step 2/6 : LABEL maintainer="Manisha Reddy"
----> Running in c40e75063875
----> Removed intermediate container c40e75063875
----> 176d7b470aad
Step 3/6 : RUN yum update -y && yum install nginx -y
----> Running in 79292a141a68
Amazon Linux 2023 repository 83 MB/s | 65 MB 00:00
Last metadata expiration check: 0:00:11 ago on Tue Jun 2 08:12:39 2026.
Dependencies resolved.
Nothing to do.
Complete!
Last metadata expiration check: 0:00:12 ago on Tue Jun 2 08:12:39 2026.
Dependencies resolved.
=====
Package Arch Version Repository Size
=====
Installing:
nginx x86_64 1:1.30.1-1.amzn2023.0.1 amazonlinux 34 k
Installing dependencies:
cracklib x86_64 2.9.6-27.amzn2023.0.2 amazonlinux 82 k
dbus x86_64 1:1.12.28-1.amzn2023.0.1 amazonlinux 8.5 k
```

```

root@ip-172-31-24-106: /home/ubuntu
libsemanage-3.4-5.amzn2023.0.2.x86_64
libunwind-1.4.0-5.amzn2023.0.3.x86_64
libutempter-1.2.1-4.amzn2023.0.2.x86_64
libxkbcommon-1.6.0-2.amzn2023.0.1.x86_64
logrotate-3.20.1-2.amzn2023.0.3.x86_64
nginx-1:1.30.1-1.amzn2023.0.1.x86_64
nginx-core-1:1.30.1-1.amzn2023.0.1.x86_64
nginx-filesystem-1:1.30.1-1.amzn2023.0.1.noarch
nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch
pam-1.5.1-8.amzn2023.0.8.x86_64
qrencode-libs-4.1.1-2.amzn2023.0.2.x86_64
shadow-utils-2:4.9-12.amzn2023.0.4.x86_64
systemd-252.23-11.amzn2023.x86_64
systemd-libs-252.23-11.amzn2023.x86_64
systemd-networkd-252.23-11.amzn2023.x86_64
systemd-pam-252.23-11.amzn2023.x86_64
systemd-resolved-252.23-11.amzn2023.x86_64
util-linux-2.37.4-1.amzn2023.0.4.x86_64
util-linux-core-2.37.4-1.amzn2023.0.4.x86_64
xkeyboard-config-2.41-1.amzn2023.0.1.noarch

Complete!
---> Removed intermediate container 79292a141a68
---> 2f5da03f409d
Step 4/6 : COPY index.html /usr/share/nginx/html/index.html
---> 7d1ae5124fbc
Step 5/6 : EXPOSE 80
---> Running in 8b7f33ca2009
---> Removed intermediate container 8b7f33ca2009
---> d7d5293a549f
Step 6/6 : CMD ["nginx", "-g", "daemon off;"]
---> Running in ad28ce227ed8
---> Removed intermediate container ad28ce227ed8
---> c5659e659eb1
Successfully built c5659e659eb1
Successfully tagged nginx-custom:v1
root@ip-172-31-24-106: /home/ubuntu#

```

docker ps

```

root@ip-172-31-24-106: /home/ubuntu# docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
amazonlinux:latest	ad882807c6c8	227MB	59.3MB	
jenkins/jenkins:lts	01c992ffef29	799MB	289MB	
nginx-custom:v1	c5659e659eb1	584MB	195MB	
nginx:latest	5aca99593157	241MB	66MB	
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB	
tomcat:latest	247650787b97	589MB	158MB	
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	

```

root@ip-172-31-24-106: /home/ubuntu#

```

→Run Container

**docker run -d **

**--name nginx-custom **

**-p 8083:80 **

nginx-custom:v1

```

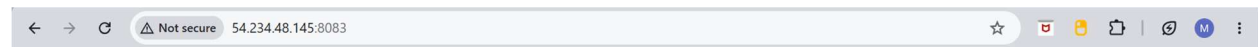
root@ip-172-31-24-106: /home/ubuntu# docker run -d \
--name nginx-custom \
-p 8083:80 \
nginx-custom:v1
d1d061050076eda82e3d2aa0fc2a1adef76a14efd6f7bbf8b42fc6e902c8f4a4
root@ip-172-31-24-106: /home/ubuntu#

```

docker ps

```
root@ip-172-31-24-106:/home/ubuntu# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS
US            PORTS
d1d061050076   nginx-custom:v1 "nginx -g 'daemon of...' 58 seconds ago Up 5
8 seconds     0.0.0.0:8083->80/tcp, [::]:8083->80/tcp
015d0be59d7b   nginx:latest   "/docker-entrypoint..." 27 minutes ago Up 2
7 minutes     0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
1f71a650f762   tomcat:latest  "catalina.sh run"        37 minutes ago Up 3
7 minutes     0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp
root@ip-172-31-24-106:/home/ubuntu#
```

→ Successfully created a custom Docker image using a Dockerfile with amazonlinux:latest, installed Nginx, and copied a custom index.html file.



Welcome to Nginx Docker Container!!

5.Push image to Docker hub

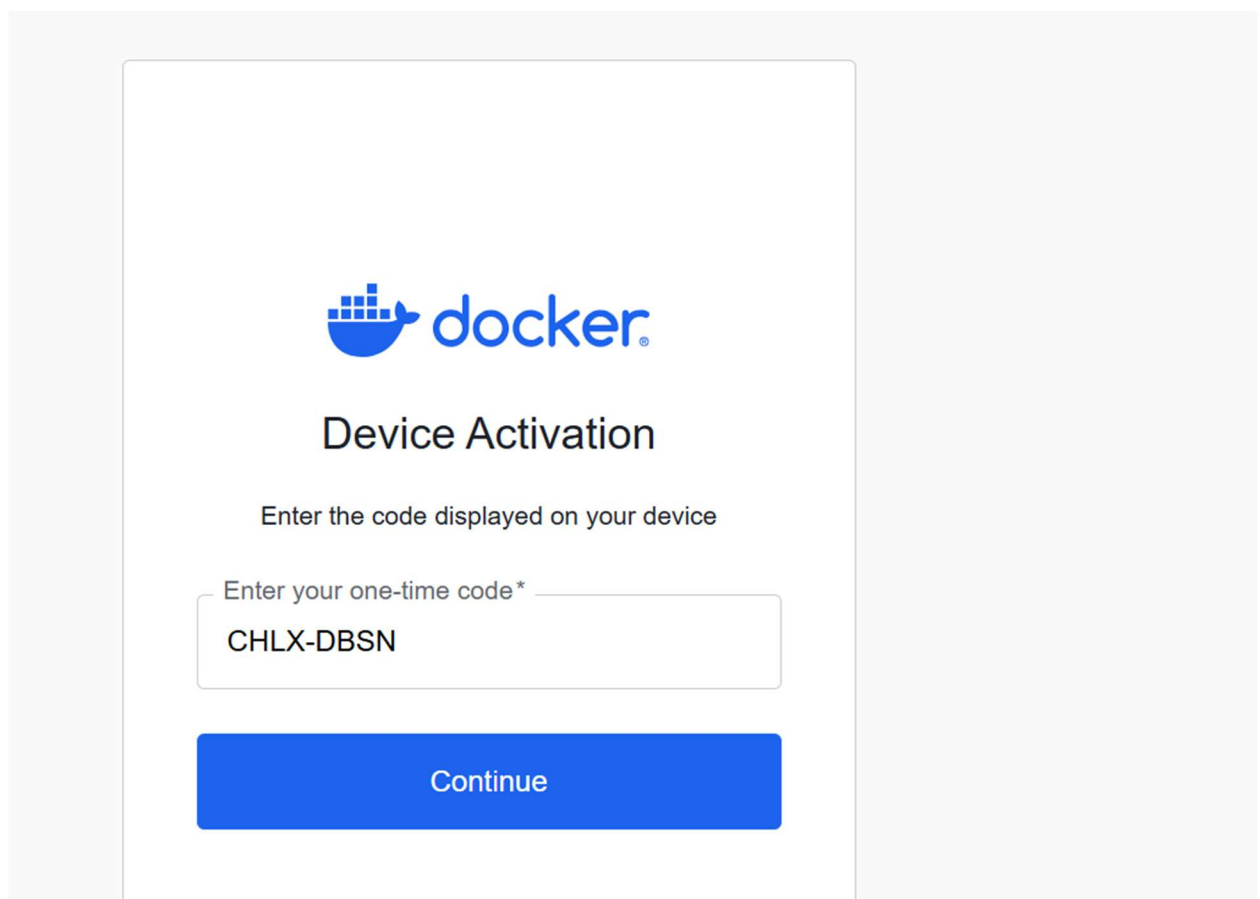
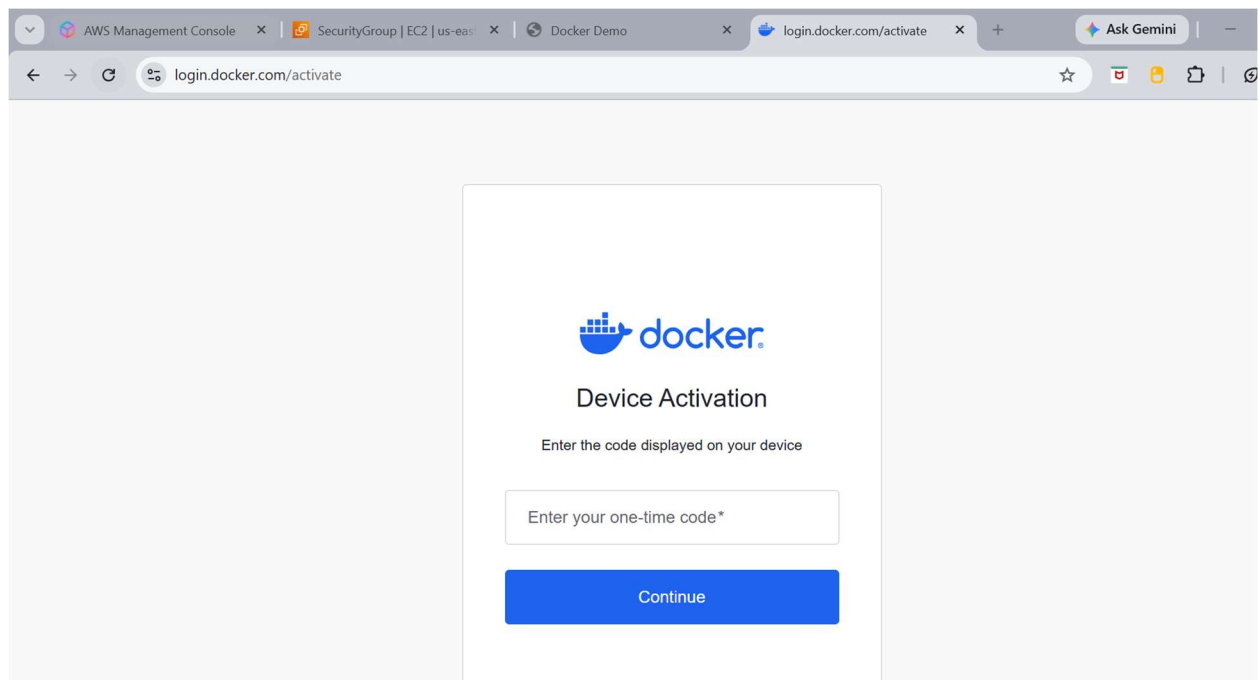
docker login

```
root@ip-172-31-24-106:/home/ubuntu
root@ip-172-31-24-106:/home/ubuntu# docker login
USING WEB-BASED LOGIN
Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: CHLX-DBSN
Press ENTER to open your browser or submit your device code here: https://login
.docker.com/activate

Waiting for authentication in the browser...
```

Open the recommended link provided and





Device Confirmation

Please confirm this is the code displayed on your
Docker CLI:

CHLX-DBSN

If you did not initiate this action or you do not
recognize this device select cancel.

Cancel

Confirm



Congratulations, you're all
set!

Your device is now connected.

```
root@ip-172-31-24-106: /home/ubuntu
root@ip-172-31-24-106: /home/ubuntu# docker login

USING WEB-BASED LOGIN

[ Info → To sign in with credentials on the command line, use 'docker login -u
<username>'

Your one-time device confirmation code is: WMDK-BBBG
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...

WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
root@ip-172-31-24-106: /home/ubuntu#
```

Create a docker account

The screenshot shows the Docker Home page for user **manisha3284**. The page has a blue header with the text "Welcome to Docker Home, manisha3284". Below the header, there are two buttons: "Get started with Docker guidance" and "Learn about Docker concepts". The left sidebar contains a navigation menu with items: Home, Hub, Build Cloud, Hardened Images, Scout, Testcontainers Cloud, Docker Desktop, Settings, and Billing. The main content area is titled "Docker products" and features three cards: "docker.desktop" (Innovate with Docker Desktop), "buildcloud" (Build with Docker Build Cloud), and "scout" (Secure with Docker Scout). Each card provides a brief description of the product's capabilities.

docker push manisha3284/nginx-custom:v1

```
root@ip-172-31-24-106:/home/ubuntu# docker push manisha3284/nginx-custom:v1
The push refers to repository [docker.io/manisha3284/nginx-custom]
e567a2894165: Pushed
c7345fb8c651: Mounted from library/amazonlinux
ce558ca2c1df: Pushed
v1: digest: sha256:c5659e659eb1989e792d8af4f7ea763d18b37caa98467610729b3c8fe1271d7c size: 972
root@ip-172-31-24-106:/home/ubuntu#
```

→Successfully Pushed image to Docker hub

The screenshot shows the Docker Hub repository page for the user **manisha3284**. The page title is "Repositories" and it states "All repositories within the manisha3284 namespace." Below the title, there is a search bar and a dropdown menu for "All content". A blue button labeled "Create a repository" is visible. The main content area displays a table of repositories. The table has columns: Name, Last Pushed, Contains, Visibility, and Scout. The table lists one repository: **manisha3284/nginx-custom**, which was pushed 1 minute ago, contains an image, is public, and has a scout status of "Inactive".

Name	Last Pushed	Contains	Visibility	Scout
manisha3284/nginx-custom	1 minute ago	IMAGE	Public	Inactive

manisha3284
Your personal account

Repositories

Hardened Images

Collaborations

Settings

Default privacy

Notifications

Billing

Usage

Pulls

Storage

Upgrade plan

Repositories / nginx-custom / General

manisha3284/nginx-custom

Last pushed 3 minutes ago · ☆ 0 · ↓ 0

Add a description

Add a category

Docker commands

To push a new tag to this repository:

```
docker push manisha3284/nginx-custom:tagname

Public view



General Tags Image Management Collaborators Webhooks Settings



Tags



DOCKER SCOUT INACTIVE  
Activate



This repository contains 1 tag(s).



| Tag | OS    | Type            | Pulled    | Pushed |
|-----|-------|-----------------|-----------|--------|
| v1  | Image | less than 1 day | 3 minutes |        |



Build with  
Docker Build Cloud



Accelerate image build times with access to cloud-based builders and shared cache.



Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated


```

6.Push image to AWS ECR

Check whether AWS CLI is installed.

Aws --version

```
root@ip-172-31-24-106:/home/ubuntu# aws --version
Command 'aws' not found, but can be installed with:
snap install aws-cli # version 1.45.18, or
apt install awscli # version 2.31.35-1
See 'snap info aws-cli' for additional versions.
root@ip-172-31-24-106:/home/ubuntu#
```

As there is no aws cli installed. Lets install

```

root@ip-172-31-24-106: /home/ubuntu
root@ip-172-31-24-106:/home/ubuntu# apt install awscli -y
Installing:
  awscli

Installing dependencies:
  docutils-common  libopenjp2-7  python3-colorama
  libdeflate0      libpaper-utils  python3-docutils
  libgraphite2-3   libpaper2      python3-olefile
  libharfbuzz0b    libraqm0       python3-pil
  libimagequant0   libsharpyuv0   python3-prompt-toolkit
  libjbig0         libtiff6       python3-roman-numerals
  libjpeg-turbo8   libwebp7       python3-ruamel.yaml
  libjpeg8         libwebpdemux2  python3-ruamel.yaml.clib
  liblcms2-2       libwebpmux3    python3-wcwidth
  liblerc4         python3-awscli

Suggested packages:
  liblcms2-utils  | ttf-linux-libertine  texlive-latex-recommended
  docutils-doc    texlive-lang-french   python-pil-doc
  fonts-linuxlibertine  texlive-latex-base

Summary:
  Upgrading: 0, Installing: 30, Removing: 0, Not Upgrading: 4
  Download size: 16.0 MB
  Space needed: 148 MB / 12.1 GB available

Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3
python3-awscli amd64 0.28.4+dfsg-1 [954 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3
python3-colorama all 0.4.6-4build1 [32.2 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 docu
tils-common all 0.22.4+dfsg-1 [130 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 pyth
on3-roman-numerals all 4.1.0-1 [8660 B]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 pyth
on3-docutils all 0.22.4+dfsg-1 [439 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 pyth
on3-wcwidth all 0.2.14+dfsg1-1build1 [26.5 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3
python3-prompt-toolkit all 3.0.52-2 [258 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3

```

Verify the version

```

root@ip-172-31-24-106:/home/ubuntu# aws --version
aws-cli/2.31.35 Python/3.14.4 Linux/7.0.0-1004-aws source/x86_64.ubuntu.26
root@ip-172-31-24-106:/home/ubuntu#

```

Configure aws cli

aws configure

```

root@ip-172-31-24-106:/home/ubuntu# aws configure
AWS Access Key ID [*****3KSQ]:
AWS Secret Access Key [*****Rx+J]:
Default region name [global]: none
Default output format [None]: json

```

aws sts get-caller-identity

```

root@ip-172-31-24-106:/home/ubuntu# aws sts get-caller-identity
{
  "UserId": "142083399660",
  "Account": "142083399660",
  "Arn": "arn:aws:iam::142083399660:root"
}
root@ip-172-31-24-106:/home/ubuntu#

```

login to aws ECR

aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 142083399660.dkr.ecr.us-east-1.amazonaws.com

```
root@ip-172-31-24-106:/home/ubuntu# aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin 142083399660.dkr.ecr.us-east-1.amazonaws.com
WARNING! Your credentials are stored unencrypted in '/root/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
root@ip-172-31-24-106:/home/ubuntu#
```

aws ecr create-repository --repository-name custom-nginx

```
root@ip-172-31-24-106:/home/ubuntu# aws ecr describe-repositories --repository-names custom-nginx --region us-east-1
{
  "repositories": [
    {
      "repositoryArn": "arn:aws:ecr:us-east-1:142083399660:repository/custom-nginx",
      "registryId": "142083399660",
      "repositoryName": "custom-nginx",
      "repositoryUri": "142083399660.dkr.ecr.us-east-1.amazonaws.com/custom-nginx",
      "createdAt": "2026-06-02T12:48:22.250000+00:00",
      "imageTagMutability": "MUTABLE",
      "imageScanningConfiguration": {
        "scanOnPush": false
      },
      "encryptionConfiguration": {
        "encryptionType": "AES256"
      }
    }
  ]
}
```

docker tag nginx-custom:v1 142083399660.dkr.ecr.us-east-1.amazonaws.com/custom-nginx:v1

verify docker images

docker images

```
root@ip-172-31-24-106:/home/ubuntu# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
142083399660.dkr.ecr.us-east-1.amazonaws.com/custom-nginx:v1	c5659e659eb1	584MB	195MB	U
443628962425.dkr.ecr.us-east-1.amazonaws.com/custom-nginx:v1	c5659e659eb1	584MB	195MB	U
amazonlinux:latest	ad882807c6c8	227MB	59.3MB	
jenkins/jenkins:lts	01c992ffef29	799MB	289MB	
manisha3284/custom-nginx:latest	c5659e659eb1	584MB	195MB	U
manisha3284/nginx-custom:v1	c5659e659eb1	584MB	195MB	U
nginx-custom:v1	c5659e659eb1	584MB	195MB	U
nginx:latest	5aca99593157	241MB	66MB	U
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB	
tomcat:latest	247650787b97	589MB	158MB	U
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	

Push the image

docker push 142083399660.dkr.ecr.us-east-1.amazonaws.com/custom-nginx:v1

```
root@ip-172-31-24-106:/home/ubuntu# docker push 142083399660.dkr.ecr.us-east-1.amazonaws.com/custom-nginx:v1
The push refers to repository [142083399660.dkr.ecr.us-east-1.amazonaws.com/custom-nginx]
e567a2894165: Pushed
ce558ca2c1df: Pushed
c7345fb8c651: Pushed
v1: digest: sha256:c5659e659eb1989e792d8af4f7ea763d18b37caa98467610729b3c8fe1271d7c size: 972
root@ip-172-31-24-106:/home/ubuntu#
```

→Pushed image successfully

us-east-1.console.aws.amazon.com/ecr/private-registry/repositories?region=us-east-1

Search

[Alt+S]

United States (N. Virginia)

Manisha A (142083399660)

Manisha A

Amazon ECR

Private registry

Repositories

Amazon Elastic Container Registry

Private registry

Repositories

Public registry

ECR public gallery

Private repositories (1)

View push commands

Delete

Actions

Create repository

Search by repository substring

	Repository name	URI	Created at	Tag immutability	Encryption type
☐	custom-nginx	142083399660.dkr.ecr.us-east-1.amazonaws.com/custom-nginx	June 02, 2026, 18:18:22 (UTC+05.5)	Mutable	AES-256

Search

[Alt+S]

United States (N. Virginia)

Manisha A (142083399660)

Manisha A

Amazon ECR

Private registry

Repositories

custom-nginx

Amazon Elastic Container Registry

Private registry

Repositories

Public registry

ECR public gallery

on ECS

custom-nginx

Summary

Images

Lifecycle policy

Permissions

Repository tags

Images (1) Info

Delete

Copy URI

Details

Scan

View push commands

Filter active images

☐	Image tags	Type	Created at	Image size (MB)	Image digest	Last pulled at
☐	v1	Image	June 02, 2026, 18:28:52 (UTC+05.5)	195.48	sha256:c...	-